Stellar Cyber 6.6.0s is now live! Discover all the updates in the release notes!

Rules Contributing to Suspicious LSASS Process Access Alert

The following rules are used to identify suspicious process access to or from the Local Security Authority Subsystem Service (LSASS). Any one or more of these will trigger the Suspicious LSASS Process Access Alert. Details for each rule can be viewed by clicking the **More Details** link in the description.

Windows Events Required: 10, 4656, 4663

The Windows Detect Profile (Low Volume) covers these required Windows events.

Title	Description

LSASS Memory Access by Tool With Dump Keyword In Name	Detects LSASS process access requests from a source process with the "dump" keyword in its image name.
	
	More details
	
	Rule ID
	`windows_process_access_1`
	
	Query
	`{'selection': {'TargetImage|endswith': '\\lsass.exe', 'SourceImage|contains': 'dump'}, 'condition': 'selection'}`
	
	Log Source
	Stellar Cyber Windows Server Sensor configured for:
	• Monitoring process access
	
	Rule Source
	SigmaHQ,9bd012ee-0dff-44d7-84a0-aa698cfd87a3
	
	Author: Florian Roth (Nextron Systems)
	
	Tactics, Techniques, and Procedures
	TA0006, T1003.001
	
	References
	• https://twitter.com/_xpn_/status/1491557187168178176

Title	Description
	https://www.ired.team/offensive-security/credential-access-and-credential-dumping/dump-credentials-from-lsass-process-without-mimikatz **Severity** 75 **Suppression Logic Based On** • computer_name • event_data.SourceImage • event_data.TargetImage • stellar.rule_id **Additional Information** **Maturity**: test **Creation Date**: 2022/02/10 **Risk Level**: high **False Positives** • Rare programs that contain the word dump in their name and access lsass

Credential Dumping Activity By Python Based Tool	Detects LSASS process access for potential credential dumping by a Python-like tool such as LaZagne or Pypykatz.
	
	More details
	
	Rule ID
	`windows_process_access_2`
	
	Query
	`{'selection': {'TargetImage|endswith': '\\lsass.exe', 'CallTrace|contains': '_ctypes.pyd'}, 'filter_av': {'SourceImage': ['?:\\Windows\\TEMP\\rapid7\\ir_agent.exe']}, 'condition': 'selection and not filter_av'}`
	
	Log Source
	
	Stellar Cyber Windows Server Sensor configured for:
	• Monitoring process access
	
	Rule Source
	
	SigmaHQ,f8be3e82-46a3-4e4e-ada5-8e538ae8b9c9
	
	Author: Bhabesh Raj, Jonhnathan Ribeiro
	
	Tactics, Techniques, and Procedures
	
	TA0006, T1003.001
	
	References
	• https://twitter.com/bh4b3sh/status/1303674603819081728
	• https://github.com/skelsec/pypykatz
	
	Severity
	
	75
	
	Suppression Logic Based On
	• computer_name
	• event_data.SourceImage
	• event_data.TargetImage
	• stellar.rule_id
	
	Additional Information
	Maturity: stable
	Creation Date: 2023/11/27
	Risk Level: high
	False Positives
	• Unknown

Title	Description
LSASS Memory Access by Process in Temp Folder	Identifies suspicious access to LSASS from a source process in Temp folder. More details **Rule ID** `windows_process_access_3` **Query** <pre>{'selection': {'TargetImage|endswith': '\\lsass.exe', 'SourceImage': ['*\\Local\\Temp*', '*\\LocalLow\\Temp*', '*\\Roaming\\Temp*']}, 'condition': 'selection'}</pre> **Log Source** Stellar Cyber Windows Server Sensor configured for: • Monitoring process access **Rule Source** Developed internally by Stellar Cyber **Tactics, Techniques, and Procedures** TA0006, T1003.001 **References** N/A **Severity** 75 **Suppression Logic Based On** • computer_name • event_data.SourceImage • event_data.TargetImage • stellar.rule_id **Additional Information** **Maturity**: test **Creation Date**: 2024/01/26 **Risk Level**: high **False Positives**: • Unknown
Suspicious LSASS Access via MalSecLogon	Identifies suspicious access to LSASS handle from a call trace pointing to seclogon.dll and with a suspicious access rights value. This may indicate an attempt to leak an LSASS handle via abusing the Secondary Logon service in preparation for credential access. More details **Rule ID** `windows_process_access_4` **Query** <pre>{'selection1': {'TargetImage|endswith': '\\lsass.exe'}, 'selection2': {'CallTrace|contains': 'seclogon.dll'}, 'selection3': {'SourceImage|endswith': 'svchost.exe'}, 'selection4': {'GrantedAccess': '0x14c0'}, 'condition': 'selection1 and selection2 and selection3 and selection4'}</pre> **Log Source** Stellar Cyber Windows Server Sensor configured for: • Monitoring process access **Rule Source**

Title	Description
	Developed internally by Stellar Cyber **Tactics, Techniques, and Procedures** TA0006, T1003.001 **References** N/A **Severity** 75 **Suppression Logic Based On** • computer_name • event_data.SourceImage • event_data.TargetImage • stellar.rule_id **Additional Information** **Maturity**: production **Creation Date**: 2022/06/29 **Risk Level**: high **False Positives**: N/A

Potential Credential Access via DuplicateHandle in LSASS	Identifies suspicious access to an LSASS handle via DuplicateHandle from an unknown call trace module. This may indicate an attempt to bypass the NtOpenProcess API to evade detection and dump LSASS memory for credential access.
	
	More details
	
	Rule ID
	
	`windows_process_access_5`
	
	Query
	
	`{'selection1': {'SourceImage|endswith': '\\lsass.exe'}, 'selection2': {'GrantedAccess': '0x40'}, 'selection3': {'CallTrace|contains': 'UNKNOWN'}, 'condition': 'selection1 and selection2 and selection3'}`
	
	Log Source
	
	Stellar Cyber Windows Server Sensor configured for:
	• Monitoring process access
	
	Rule Source
	
	Developed internally by Stellar Cyber
	
	Tactics, Techniques, and Procedures
	
	TA0006, T1003.001
	
	References
	N/A
	
	Severity
	
	50
	
	Suppression Logic Based On
	• computer_name
	• event_data.SourceImage
	• event_data.TargetImage
	• stellar.rule_id
	
	Additional Information
	Maturity: production
	Creation Date: 2021/09/27
	Risk Level: medium
	False Positives: N/A

Title	Description

Potential Credential Access via LSASS Memory Dump	Identifies suspicious access to LSASS handle from a call trace pointing to DBGHelp.dll or DBGCore.dll, which both export the MiniDumpWriteDump method that can be used to dump LSASS memory content in preparation for credential access.
	
	More details
	
	Rule ID
	`windows_process_access_6`
	
	Query
	`{'selection1': {'TargetImage|endswith': '\\lsass.exe'}, 'selection2': {'CallTrace|contains': ['dbghelp.dll', 'dbgcore.dll']}, 'selection3': {'SourceImage': ['?:\\Windows\\System32\\WerFault.exe', '?:\\Windows\\System32\\WerFaultSecure.exe', '?:\\Windows\\System32\\tasklist.exe']}, 'condition': 'selection1 and selection2 and (not selection3)'}`
	
	Log Source
	Stellar Cyber Windows Server Sensor configured for:
	• Monitoring process access
	
	Rule Source
	Developed internally by Stellar Cyber
	
	Tactics, Techniques, and Procedures
	TA0006, T1003.001
	
	References
	N/A
	
	Severity
	75
	
	Suppression Logic Based On
	• computer_name
	• event_data.SourceImage
	• event_data.TargetImage
	• stellar.rule_id
	
	Additional Information
	Maturity: production; Creation Date: 2021/10/07; Risk Level: high; False Positives: N/A

Password Dumper Activity on LSASS	Detects process handle on LSASS process with certain access mask and object type SAM_DOMAIN
	
	More details
	
	Rule ID
	`windows_security_146`
	
	Query
	`{'selection': {'EventID': 4656, 'ProcessName|endswith': '\\lsass.exe', 'AccessMask': '0x705', 'ObjectType': 'SAM_DOMAIN'}, 'condition': 'selection'}`
	
	Log Source
	Stellar Cyber Windows Server Sensor configured for:
	• Collecting Windows security events

Title	Description
	Resource SigmaHQ,aa1697b7-d611-4f9a-9cb2-5125b4ccfd5c Author: sigma **Tactics, Techniques, and Procedures** TA0006, T1003.001 **References** • https://twitter.com/jackcr/status/807385668833968128 **Severity** 75 **Suppression Logic Based On** • computer_name • event_data.ObjectName • stellar.rule_id **Additional Information** Maturity: test Creation Date: 2017/02/12 Risk Level: high False Positives: • Unknown

Potentially Suspicious AccessMask Requested From LSASS	Detects process handle on LSASS process with certain access mask
	More details
	Rule ID
	`windows_security_169`
	Query
	`{'selection_1': {'EventID': 4656, 'ObjectName|endswith': '\\lsass.exe', 'AccessMask|contains': ['0x40', '0x1400', '0x100000', '0x1410', '0x1010', '0x1438', '0x143a', '0x1418', '0x1f0fff', '0x1f1fff', '0x1f2fff', '0x1f3fff']}, 'selection_2': {'EventID': 4663, 'ObjectName|endswith': '\\lsass.exe', 'AccessList|contains': ['4484', '4416']}, 'filter_main_specific': {'ProcessName|endswith': ['\\csrss.exe', '\\GamingServices.exe', '\\lsm.exe', '\\MicrosoftEdgeUpdate.exe', '\\minionhost.exe', '\\MRT.exe', '\\MsMpEng.exe', '\\perfmon.exe', '\\procexp.exe', '\\procexp64.exe', '\\svchost.exe', '\\taskmgr.exe', '\\thor.exe', '\\thor64.exe', '\\vmtoolsd.exe', '\\VsTskMgr.exe', '\\wininit.exe', '\\wmiprvse.exe', '\\WmiPrvSE.exe', 'RtkAudUService64'], 'ProcessName|contains': [':\\Program Files (x86)\\', ':\\Program Files\\', ':\\ProgramData\\Microsoft\\Windows Defender\\Platform\\', ':\\Windows\\SysNative\\', ':\\Windows\\System32\\', ':\\Windows\\SysWow64\\', ':\\Windows\\Temp\\asgard2-agent\\']}, 'filter_main_generic': {'ProcessName|contains': ':\\Program Files'}, 'filter_main_exact': {'ProcessName|endswith': [':\\Windows\\System32\\taskhostw.exe', ':\\Windows\\System32\\msiexec.exe', ':\\Windows\\CCM\\CcmExec.exe', '\\Windows\\explorer.exe', '\\jre\\bin\\java.exe', ':\\Windows\\LTSvc\\LTSVC.exe']}, 'filter_main_sysmon': {'ProcessName|endswith': ':\\Windows\\Sysmon64.exe', 'AccessList|contains': '%%4484'}, 'filter_main_aurora': {'ProcessName|contains': ':\\Windows\\Temp\\asgard2-agent-sc\\aurora\\', 'ProcessName|endswith': '\\aurora-agent-64.exe', 'AccessList|contains': '%%4484'}, 'filter_main_scenarioengine': {'ProcessName|endswith': '\\x64\\SCENARIOENGI``

Title	Description


```
NI    E', 'AccessList|contains': '%%4484'}, 'filter_main_avi
ra1': {'ProcessName|contains|all': [':\\Users\\', '\\AppDat
a\\Local\\Temp\\is-
'], 'ProcessName|endswith': '\\avira_system_speedup.tmp', '
AccessList|contains': '%%4484'}, 'filter_main_avira2': {'Pr
ocessName|contains': ':\\Windows\\Temp\\', 'ProcessName|end
swith': '\\avira_speedup_setup_update.tmp', 'AccessList|con
tains': '%%4484'}, 'filter_main_snmp': {'ProcessName|endswi
th': ':\\Windows\\System32\\snmp.exe', 'AccessList|contains
': '%%4484'}, 'filter_main_googleupdate': {'ProcessName|con
tains': ':\\Windows\\SystemTemp\\', 'ProcessName|endswith':
 '\\GoogleUpdate.exe', 'AccessList|contains': '%%4484'}, 'f
ilter_optional_procmon': {'ProcessName|endswith': ['\\procm
on64.exe', '\\procmon.exe'], 'AccessList|contains': '%%4484
'}, 'condition': '1 of selection_* and not 1 of filter_main
_* and not 1 of filter_optional_*'}
```


Log Source

Stellar Cyber Windows Server Sensor configured for:

- Collecting Windows security events

Rule Source

SigmaHQ,4a1b6da0-d94f-4fc3-98fc-2d9cb9e5ee76

Author: Roberto Rodriguez, Teymur Kheirkhabarov, Dimitrios Slamaris, Mark Russinovich, Aleksey Potapov, oscd.community (update)

Tactics, Techniques, and Procedures

TA0006, T1003.001

References

- https://web.archive.org/web/20230208123920/https://cyberwardog.blogspot.com/2017/03/chronicles-of-threat-hunter-hunting-for_22.html
- https://www.slideshare.net/heirhabarov/hunting-for-credentials-dumping-in-windows-environment

Severity

50

Suppression Logic Based On

- computer_name
- event_data.ObjectName
- stellar.rule_id

Additional Information

Maturity	Creation Date	Risk Level	False Positives
experimental	2019/11/01	medium	• Legitimate software accessing LSASS process for legitimate reason; update the whitelist with it

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